Tree

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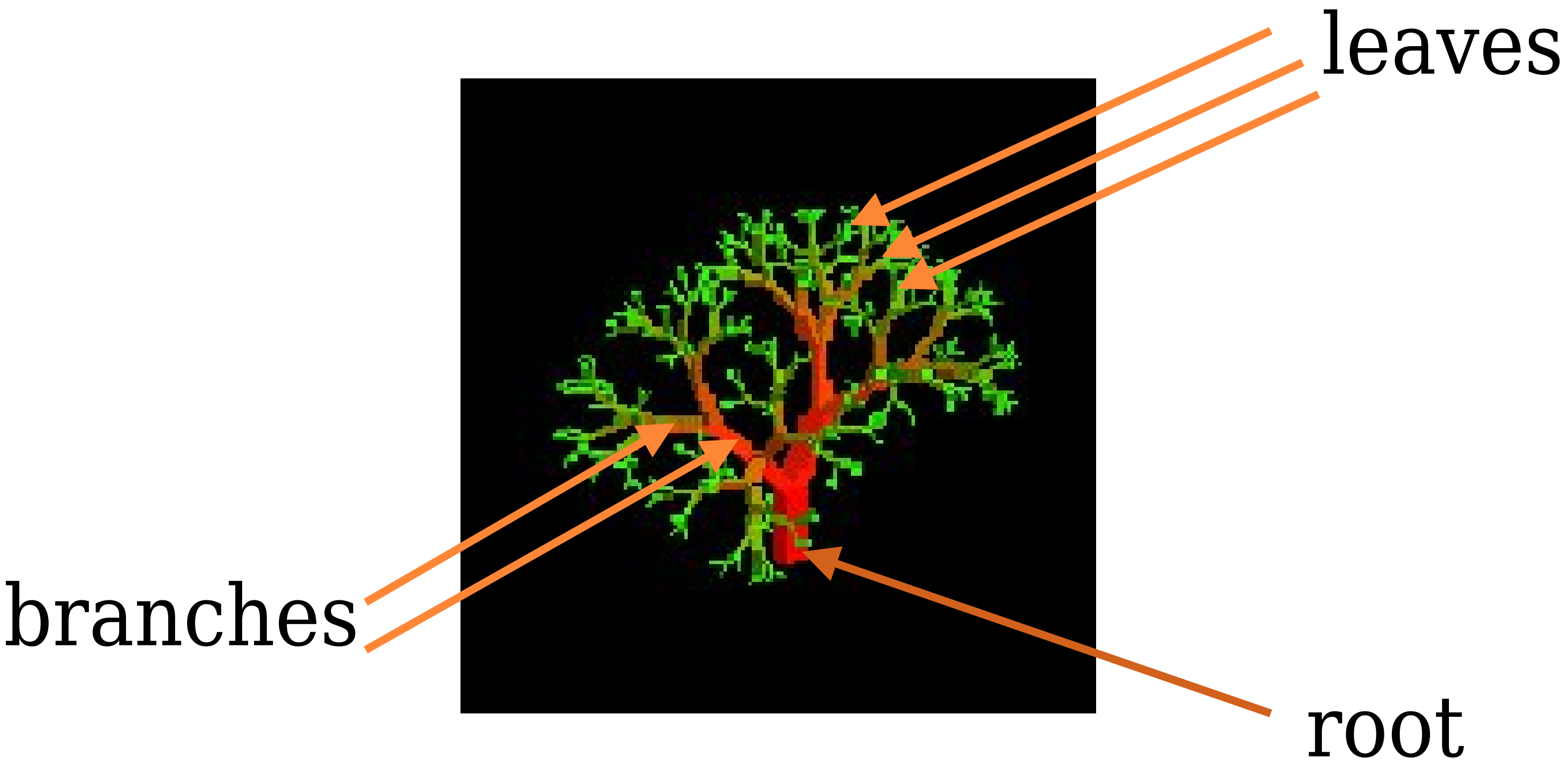
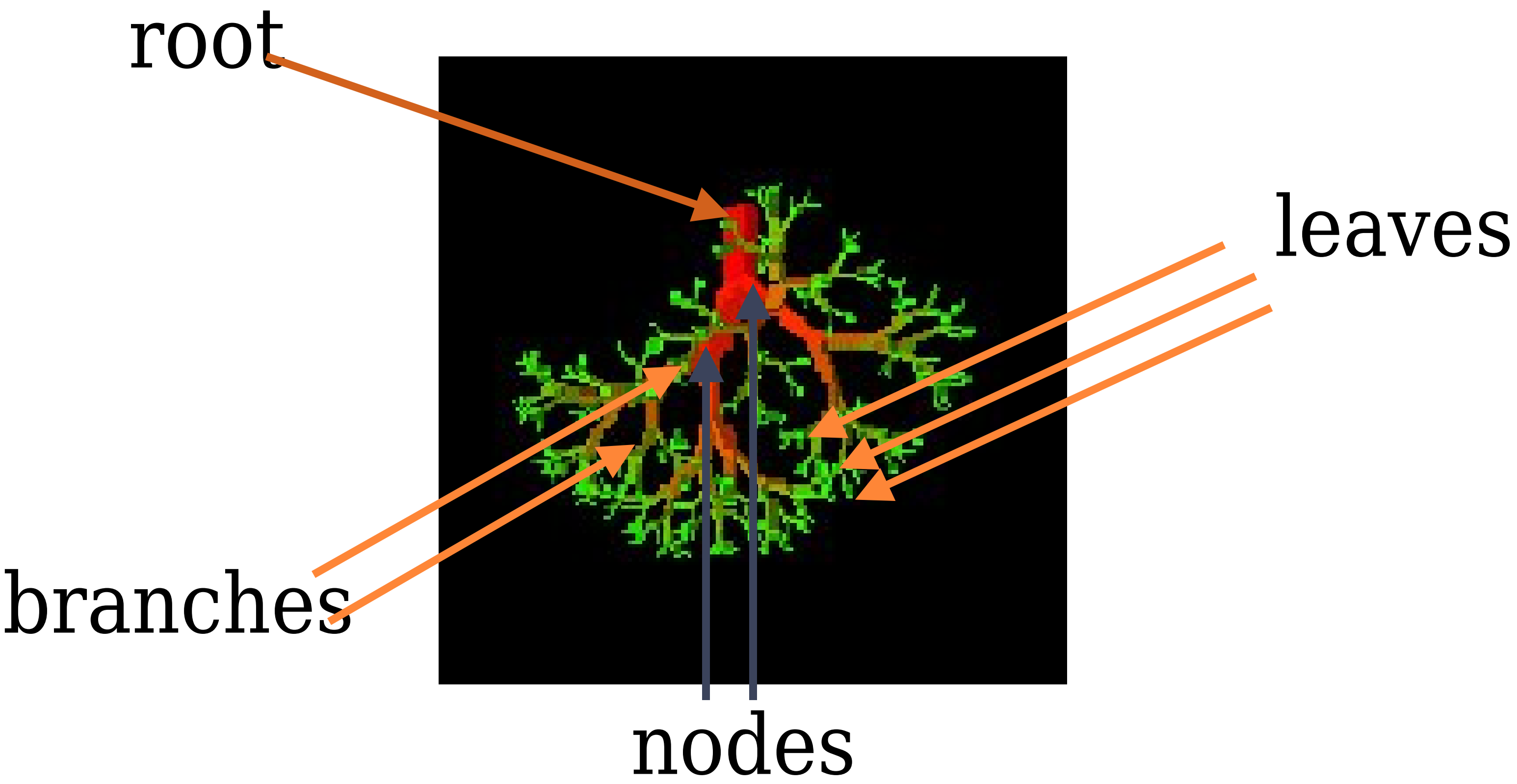
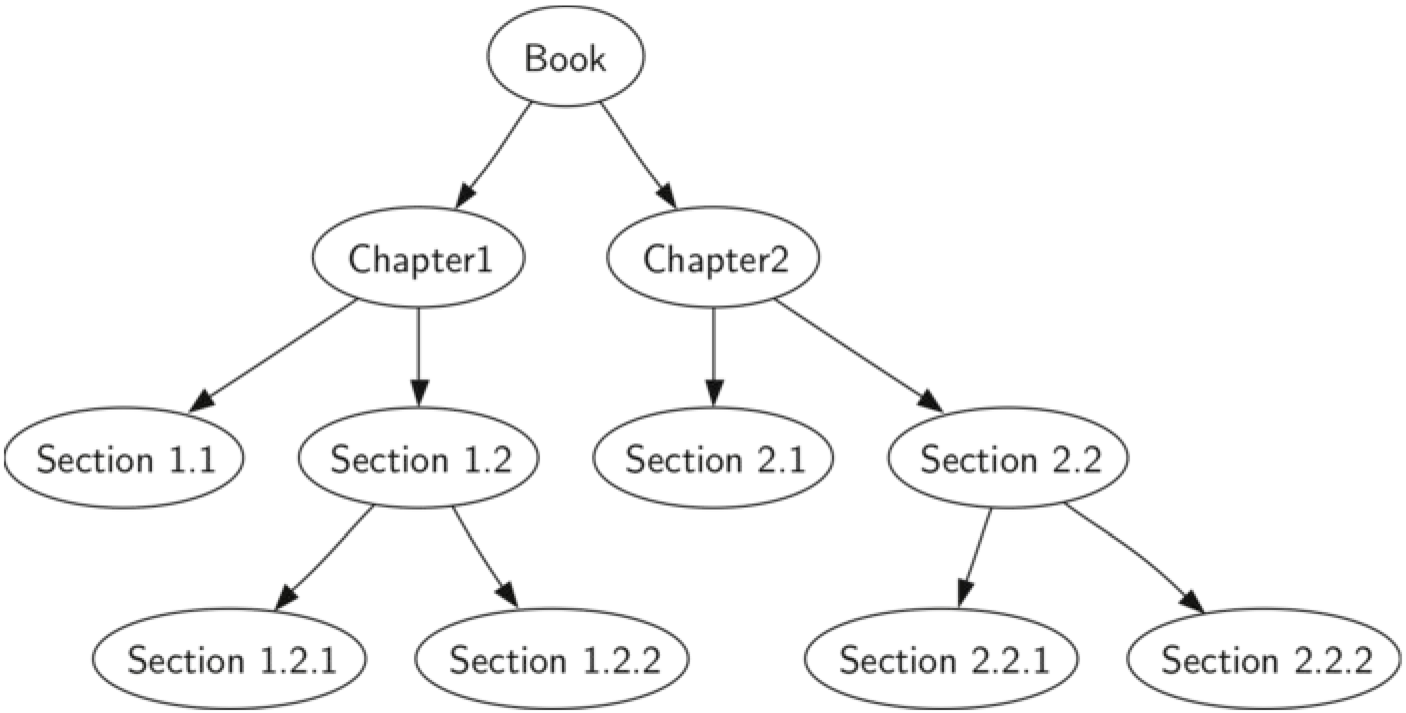
Nature View of a Tree



COMPUTER SCIENTIST'S VIEW



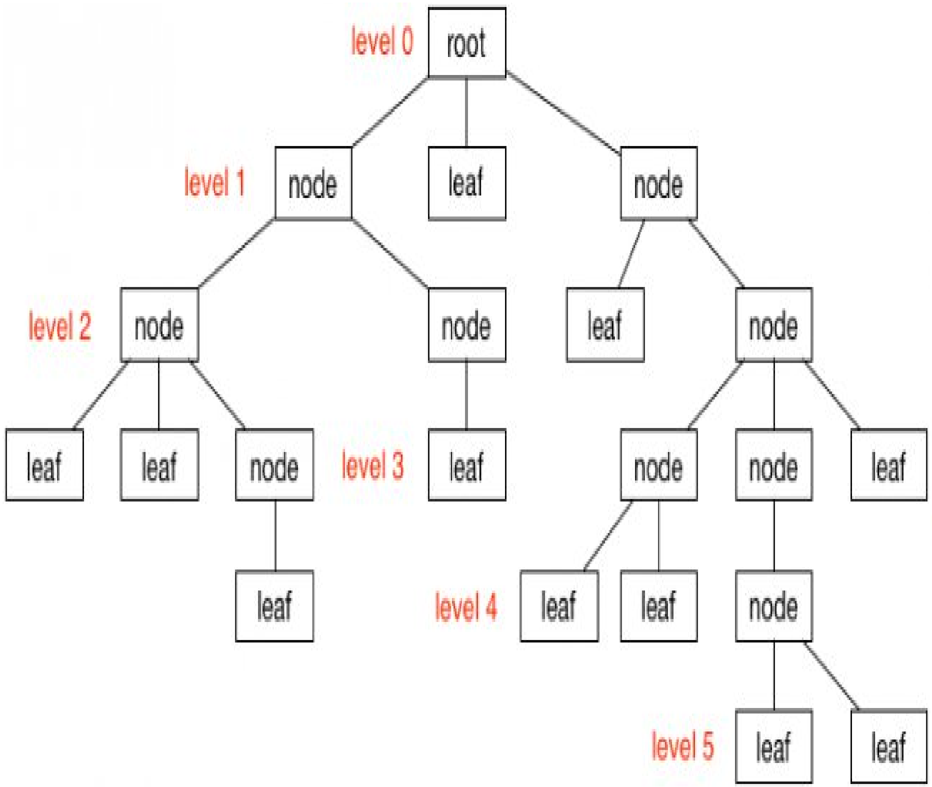
Data Structure tree

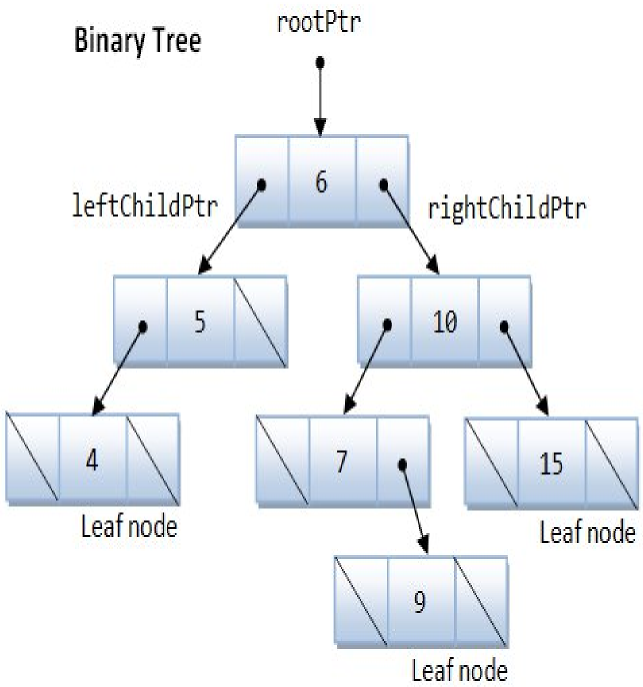


Definition of Tree

- A tree is a non empty finite set of one or more nodes such that:
- There is a specially designated node called the root.
- The remaining nodes are partitioned into n>=0 disjoint sets T_1, ..., T_n, where each of these sets can be a tree.

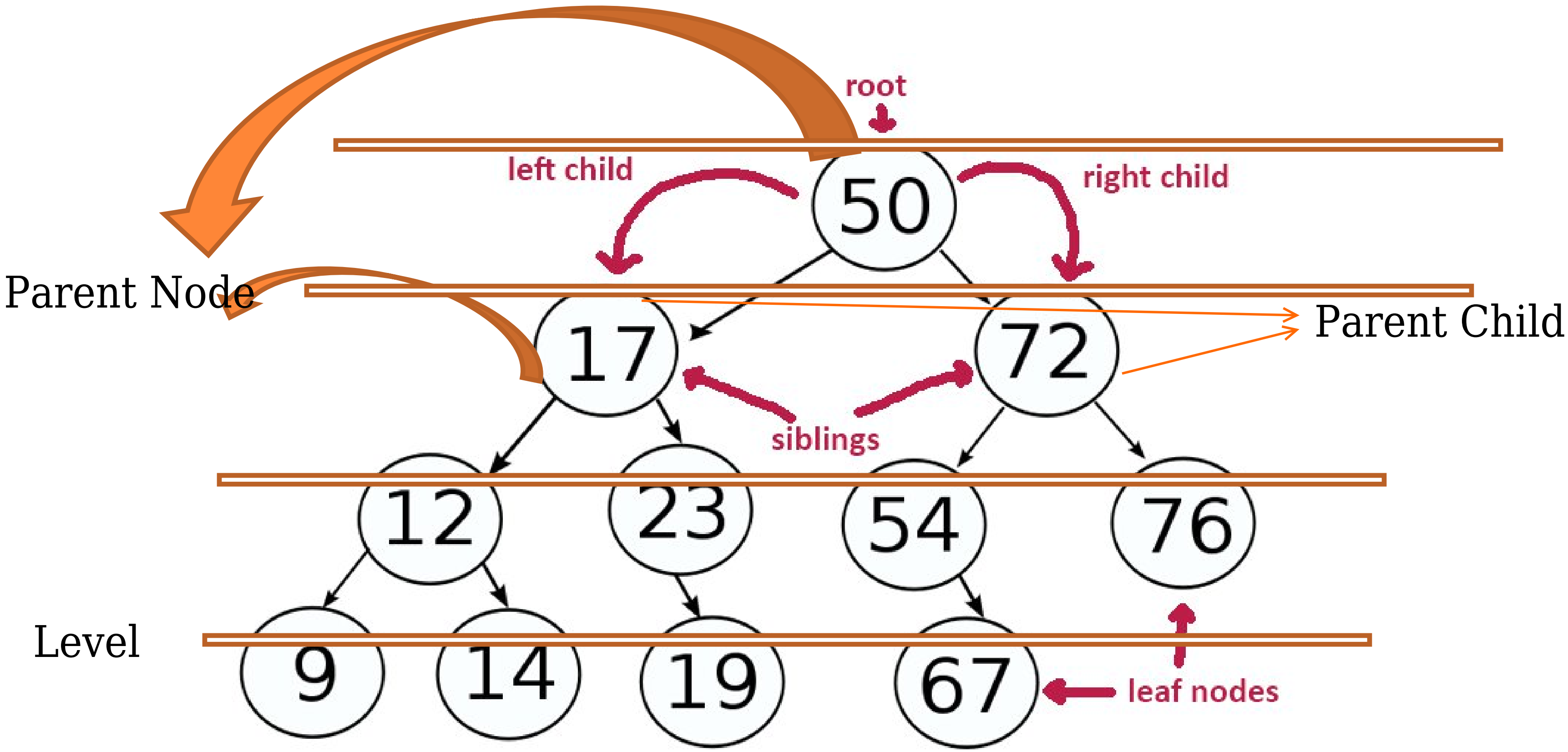
EXAMPLES





Afsana begum, Lecturer, SWE, DIU

VARIOUS TERMS OF A TREE



Terminology

Root: If the tree is not empty, the first node is called the root

Degree: Connection with nodes by edges

- Indegree: Total no of incoming edges
- Outdegree: Total no of outgoing edges

Parent Node: A node is a parent if it has successor nodes—that is, if it has an outdegree greater than zero.

Child Node: a node with a predecessor is a child. A child nodehas an indegree of one.

Leaf Node: A leaf is any node with an outdegree of zero, that is, a node with no suc-cessors.

Sibling: Two or more nodes with the same parent are siblings

Terminology

Path: A path is a sequence of nodes in which each node is adjacent to the next one.

Level: The level of a node is its distance from the root. Because the root has a zero distance from itself, the root is at level 0.

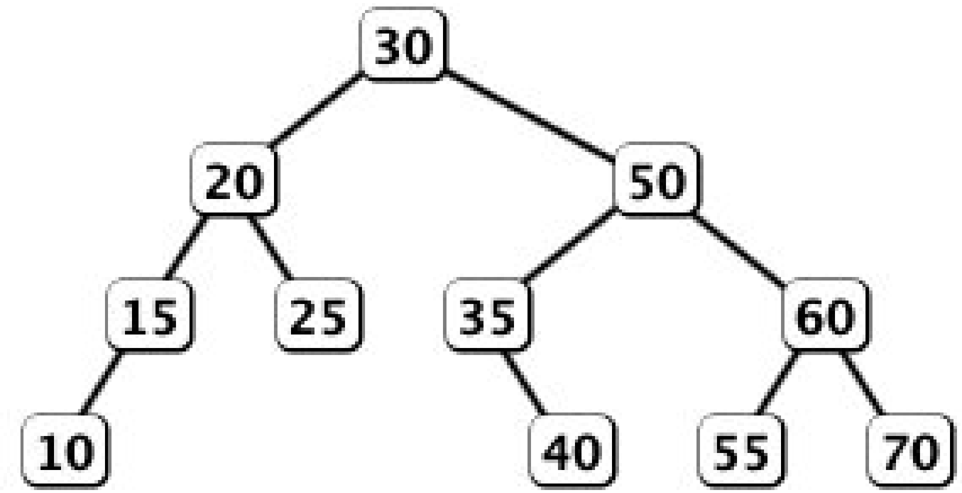
Height :The height of the tree is the level (starts from 0) of the leaf in the longest path from the root plus 1

Depth: (Similar as Height, just node to root)

Size : Total Number of Elements/Nodes

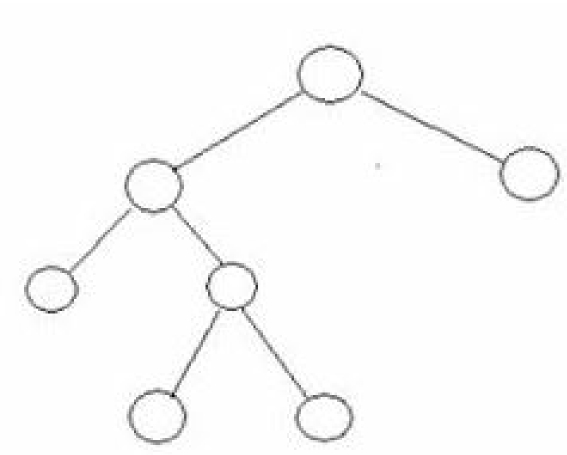
Binary Tree

Binary tree is a tree data structure in which each node has at most two children, which are referred to as the *left child* and the *right child*.



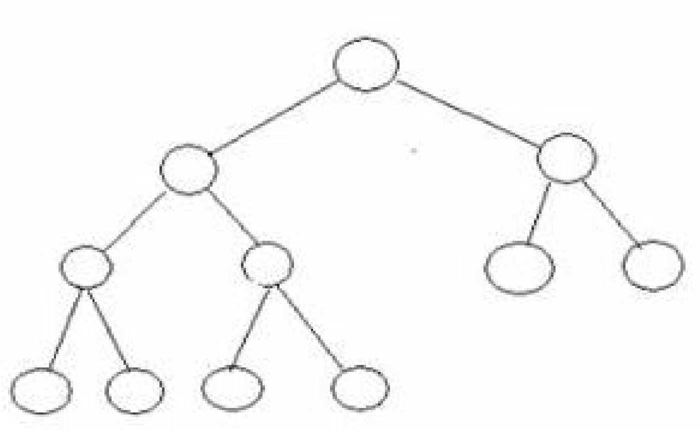
FULL TREE

A binary tree in which each node has exactly zero or two children is called **a full binary tree**. In a full tree, there are no nodes with exactly one child.



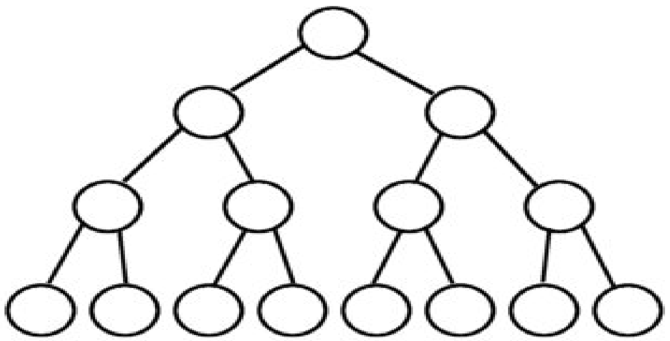
Complete Tree

A Complete Binary Tree is a tree, which is completely filled, with the possible exception of the bottom level, which is filled from left to right. A complete binary tree of the height h has between 2^h and $2^{(h+1)}$-1 nodes. Here are some examples, where h=3:



Nearly Complete Binary Tree

Full Binary Tree



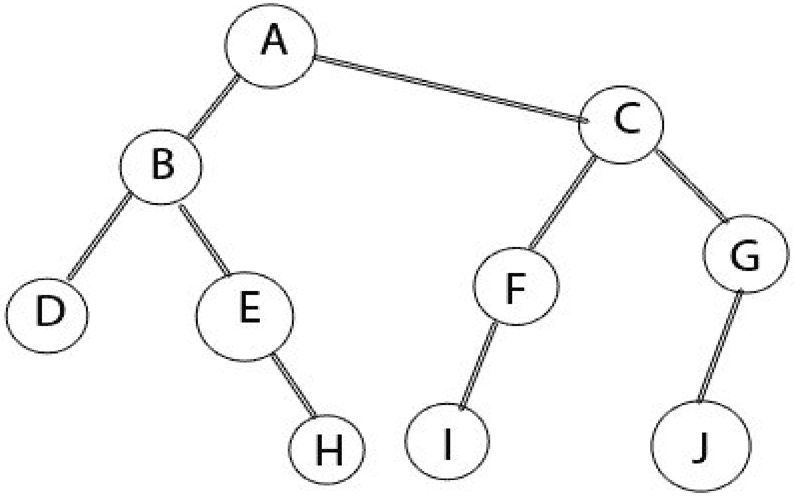
Tree Traverse

Pre Order Tree (Root, Left Child, Right Child)

In Order Tree (Left Child, Root, Right Child)

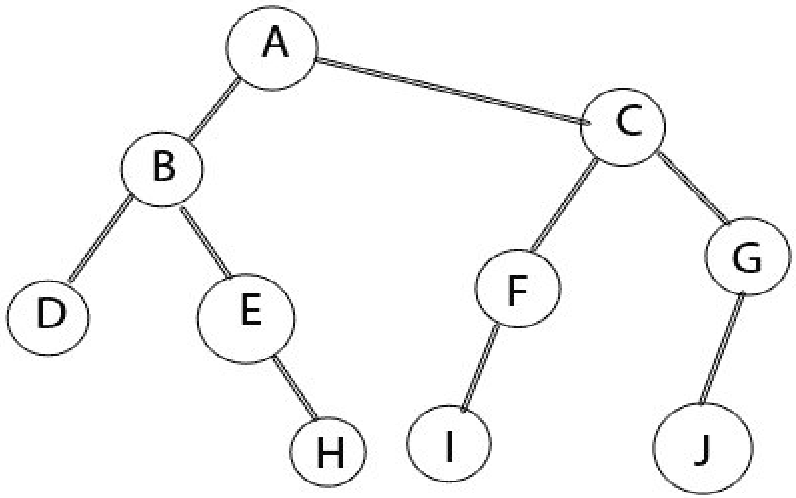
Post Order Tree (Left Child, Right Child , Root)

Pre Order (ro,L,R)



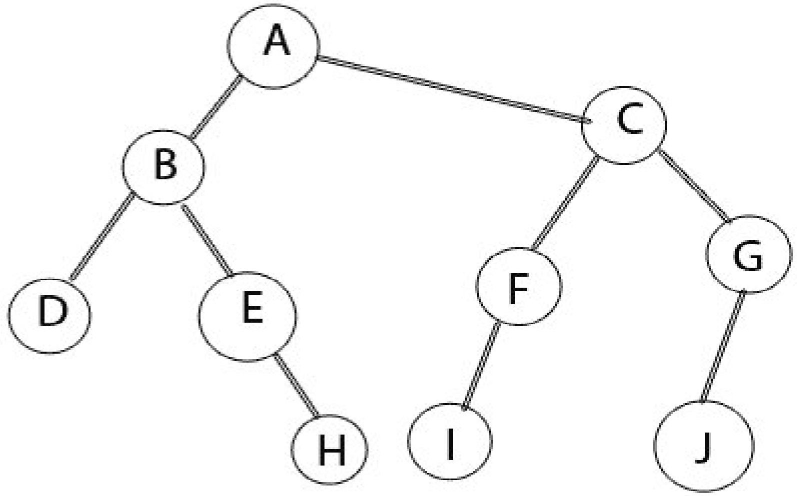
Pre Order: A,B,D,E,H,C,F,I,G,J

In Order (L,Ro,R)



In Order: D,B,E,H,A,I,F,C,J,G

POST ORDER (L, R, RO)



Post Order: D,H,E,B,I,F,J,G,C,A

Thank You

Any Question?